

Problem Set 5: Solution

1. Butterworth magnitude response

For an $N = 4$ analog Butterworth lowpass filter,

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}.$$

Compute $|H_c(j\Omega)|$ in dB at (i) $\Omega = \Omega_c$ and (ii) $\Omega = 2\Omega_c$.

$$N=4 \quad |H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^8}$$

$$(i) \quad \Omega = \Omega_c \Rightarrow |H_c(j\Omega_c)|^2 = \frac{1}{2}$$

$$|H_c(j\Omega_c)| = \frac{1}{\sqrt{2}} \quad 20 \log_{10} \frac{1}{\sqrt{2}} = -3.01 \text{ dB}$$

$$(ii) \quad \Omega = 2\Omega_c \Rightarrow |H_c(j2\Omega_c)|^2 = \frac{1}{1+2^8} = \frac{1}{257}$$

$$|H_c(j2\Omega_c)| = 0.0624$$

$$20 \log_{10} 0.0624 = -24.1 \text{ dB}$$

2. Butterworth order and cutoff behavior

The Butterworth magnitude-squared response is

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}.$$

(a) Derive the standard Butterworth low-pass order formula that guarantees a passband attenuation $\leq A_p$ in dB at Ω_p in rad/sec and a stopband attenuation $\geq A_s$ in dB at Ω_s in rad/sec:

$$N \geq \frac{\log_{10} \left[\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1} \right]}{2 \log_{10} (\Omega_s/\Omega_p)}$$

Hints: Write $|H_c(j\Omega_p)|^2$ and $|H_c(j\Omega_s)|^2$ in terms of N , form a ratio, and solve for N .

(b) For $A_p = 1$ dB at $\Omega_p = 1$ rad/sec and $A_s = 40$ dB at $\Omega_s = 3$ rad/sec, compute the minimum required order N . Round up to the nearest integer and explain why you round up rather than down.

(c) If you change Ω_s in part (b) to $\Omega_s = 2$ rad/sec, a 33% change in Ω_s , how does N change?

$$(a) \quad \frac{1}{1 + (\Omega_p/\Omega_c)^{2N}} \geq 10^{-A_p/10}$$

$$\Rightarrow 10^{A_p/10} \geq 1 + (\Omega_p/\Omega_c)^{2N} \quad \text{or}$$

$$(\Omega_p/\Omega_c)^{2N} \leq 10^{A_p/10} - 1$$

$$\Rightarrow \frac{1}{(\Omega_p/\Omega_c)^{2N}} \geq \frac{1}{10^{A_p/10} - 1}$$

we want attenuation less than A_p in passband

$$\frac{1}{1 + (\Omega_s/\Omega_c)^{2N}} < 10^{-A_s/10}$$

$$\Rightarrow 10^{A_s/10} < 1 + (\Omega_s/\Omega_c)^{2N} \quad \text{or}$$

$$(\Omega_s/\Omega_c)^{2N} > 10^{A_s/10} - 1$$

we want attenuation by more than A_s in stopband

$$\Rightarrow \frac{(\Omega_s/\Omega_c)^{2N}}{(\Omega_p/\Omega_c)^{2N}} = \left(\frac{\Omega_s}{\Omega_p} \right)^{2N} \geq \frac{10^{A_s/10} - 1}{10^{A_p/10} - 1}$$

Taking $\log_{10}$

$$2N \log_{10} \left(\frac{\Omega_s}{\Omega_p} \right) \geq \log_{10} \left(\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1} \right)$$

$$\Rightarrow N \geq \frac{\log_{10} \left(\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1} \right)}{2 \log_{10} \left(\frac{\Omega_s}{\Omega_p} \right)}$$

(b) $A_p = 1$ dB at $\Omega_p = 1$ rad/sec and

$A_s = 40$ dB at $\Omega_s = 3$ rad/sec

$$\Rightarrow N \geq \frac{\log_{10} \left(\frac{10^4 - 1}{10^{0.1} - 1} \right)}{2 \log_{10}(3)} = 4.8067 \Rightarrow N = 5$$

Rounding up to satisfy both inequalities, the one for attenuation by less than A_p in the passband, and the one for more attenuation than A_s in the stop-band.

(c) $\Omega_s = 2$ 33% decrease

$$\Rightarrow N \geq \frac{\log_{10} \left(\frac{10^4 - 1}{10^{0.1} - 1} \right)}{2 \log(2)} = 7.618 \Rightarrow N = 8$$

60% increase

3. Stability under the bilinear transform

The bilinear transform is

$$z = \frac{1 + sT/2}{1 - sT/2}$$

(a) For $T = 0.1$ sec and analog pole

$$s_p = -2 + j3,$$

compute the mapped digital pole and verify that it lies inside the unit circle.

(b) Determine the warped digital frequency ω from the analog frequency Ω using

$$\omega = 2 \tan^{-1}(\Omega T/2)$$

and verify your result in (a).

$$\begin{aligned} \text{(a)} \quad z_p &= \frac{1 + 0.05(-2 + j3)}{1 - 0.05(-2 + j3)} = \frac{0.9 + j0.15}{1.1 - j0.15} \\ &= \frac{(1.1 + j0.15)(0.9 + j0.15)}{(1.1 + j0.15)(1.1 - j0.15)} = \frac{0.9675 + j0.3}{1.2325} \\ &= 0.785 + j0.2434 \Rightarrow |z_p| = 0.8219 < 1 \end{aligned}$$

$$\text{(b)} \quad \Omega = 3 \text{ rad/sec,}$$

$$\omega = \tan^{-1}\left(\frac{0.2434}{0.785}\right) = 0.301 \text{ rad/sample}$$

$$2 \tan^{-1}\left(\frac{\Omega T}{2}\right) = 2 \tan^{-1}\left(\frac{0.3}{2}\right) = 0.298 \approx \omega$$

check also

$$\frac{2}{T} \tan\left(\frac{\omega}{2}\right) = 20 \tan\left(\frac{0.3}{2}\right) = 3.023 \approx \Omega$$

4. Prewarping design exercise

You need a digital Butterworth low-pass filter with $T = 1$ (i.e., $f_s = 1$ Hz, normalized) and a digital passband edge at $\omega_p = 0.3\pi$ where the response should be at -1 dB, and a digital stopband edge at $\omega_s = 0.6\pi$ where the response should be at least -30 dB down.

(a) Prewarp both edges to analog frequencies Ω_p and Ω_s using

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right).$$

(b) Using your formula from problem 2, compute the required Butterworth order N .

(c) In Python, design the analog Butterworth prototype at the prewarped Ω_p (adjusted to Ω_c , comment on why and how you did it) using `scipy.signal.butter(N, Wc, analog=True)`, then apply `scipy.signal.bilinear` (or `bilinear_zpk`) to obtain the digital filter. Plot and label the digital magnitude response and confirm the -1 dB and -30 dB points land at (approximately) 0.3π and 0.6π .

(d) Compute, plot and label the group delay $\tau_g(\omega)$ over $0 \leq \omega \leq \pi$ using `scipy.signal.group_delay`.

(e) Repeat the design in (c) without prewarping (i.e., using $\Omega_p = \omega_p$, $\Omega_s = \omega_s$ directly as analog frequencies) and show on the same plot how far the resulting digital cutoff misses the intended target. Quantify the error in normalized frequency.

$$T=1, \omega_p=0.3\pi, A_p=1 \text{ dB}, \omega_s=0.6\pi, A_s=30 \text{ dB}$$

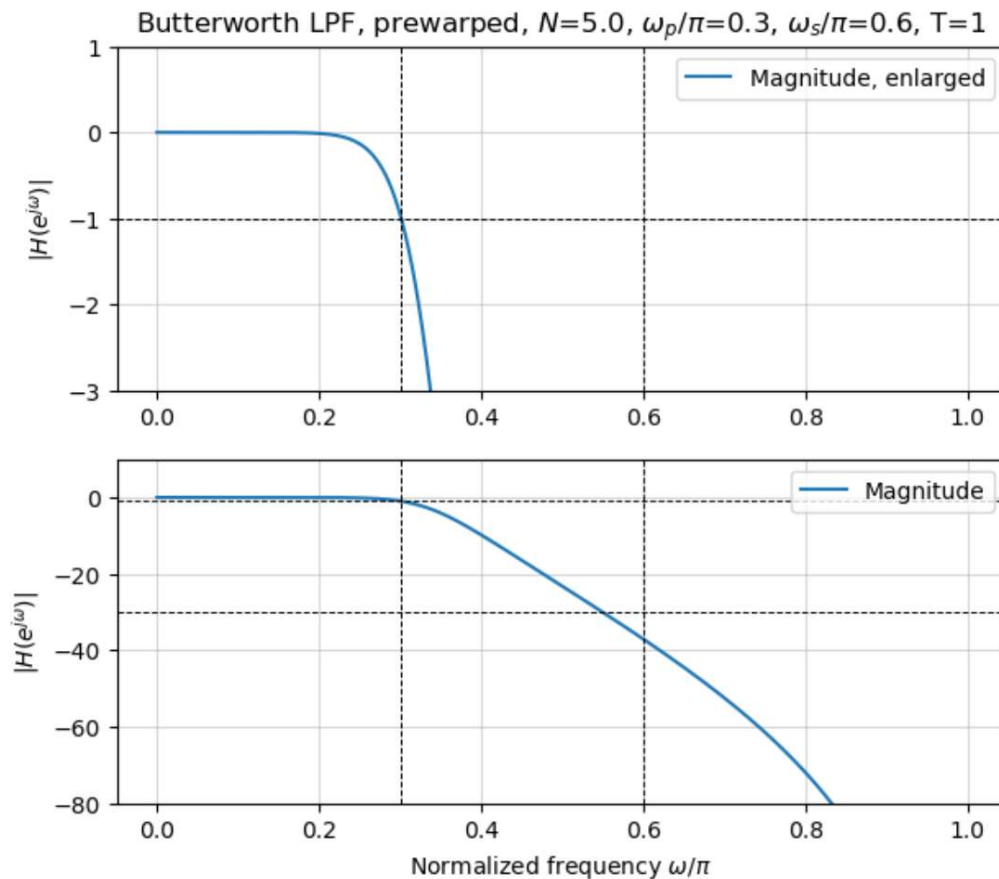
$$(a) \Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right) = 2 \tan(0.15\pi) = 1.019 \text{ rad/s}$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) = 2 \tan(0.3\pi) = 2.753 \text{ rad/s}$$

$$(b) N \geq \frac{\log_{10}\left(\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1}\right)}{2 \log_{10}(\Omega_s/\Omega_p)} = \frac{\log_{10}\left(\frac{10^3 - 1}{10^{0.1} - 1}\right)}{2 \log_{10}(2.753/1.019)} \\ = 4.155 \Rightarrow \text{use } N = 5$$

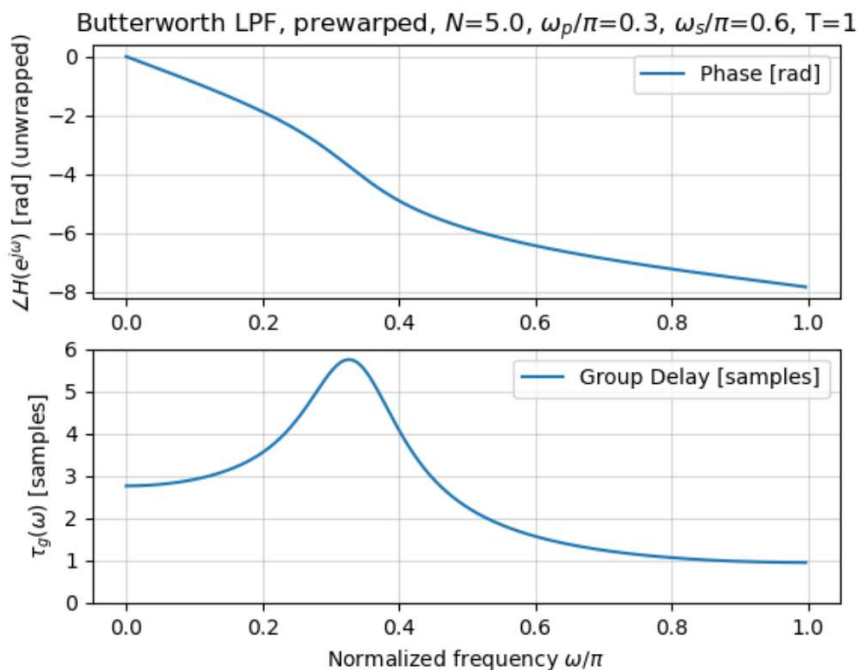
(c) `scipy.signal.butter` uses Ω_c based on -3 dB corner frequency $\Rightarrow$ need to adjust Ω_c for desired Ω_p by setting $\Omega_c \approx 1.15 \Omega_p$

(c) (cont'd.) See Python code in Appendix



$\Omega_c = 1.15 \Omega_p$
adjusted
for match
at ω_p

$N=5$
exceeds
 A_s at ω_s
 $N=4$ would
attenuate
too little



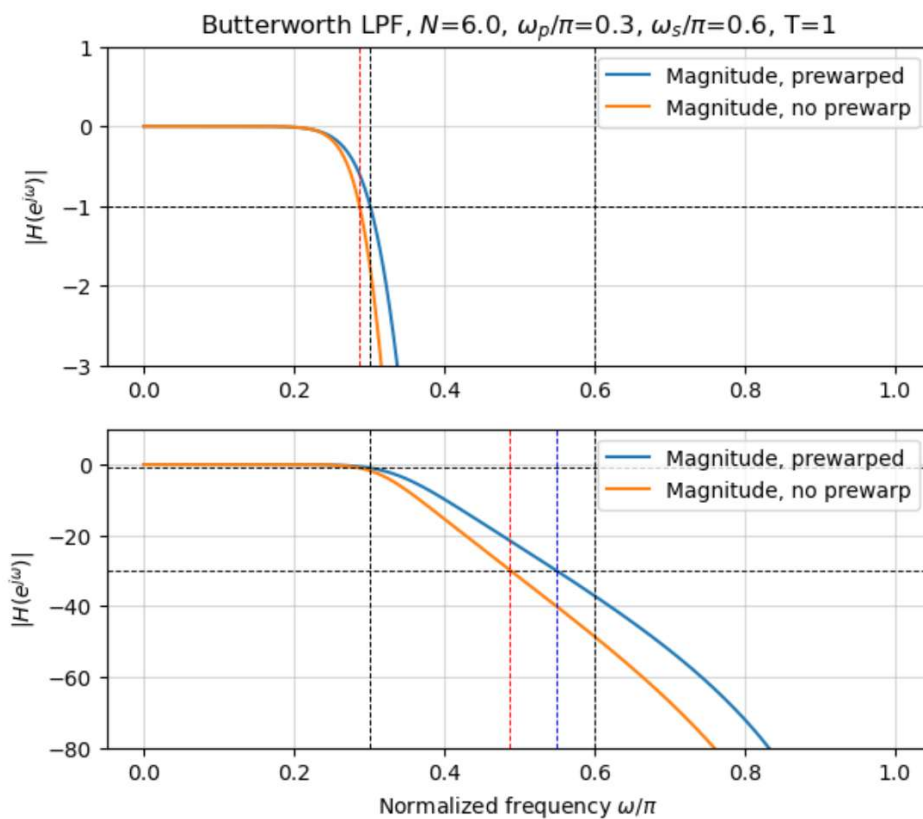
(d) Phase and
group delay.

Note peak near
 $\omega_s = 0.3\pi$

(e) Without prewarping $\Omega_p = \omega_p = 0.3\pi$
and $\Omega_s = \omega_s = 0.6\pi$

$$N \geq \frac{\log_{10}\left(\frac{10^3 - 1}{10^{0.1} - 1}\right)}{2 \log_{10}\left(\frac{0.6}{0.3}\right)} = 5.957 \Rightarrow N = 6$$

Keeping Ω_c at $\Omega_c = 1.15 \cdot \Omega_p$.



ω_p goes to
 $0.995 \omega_p$, a
-0.5% change

ω_s (at -30 dB)
goes from
 $0.915 \omega_s$ to
 $0.81 \omega_s$, a
-11.5% change

5. Window method as frequency-domain convolution (ECEN 5632 Problem)

The window method result $H_w(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) * W(e^{j\omega})$ is a periodic convolution.

(a) Explain intuitively why a *narrower* main lobe of $W(e^{j\omega})$ leads to a *sharper* transition band in $H_w(e^{j\omega})$.

(b) Explain intuitively why *lower* sidelobes in $W(e^{j\omega})$ lead to *lower* ripple in $H_w(e^{j\omega})$, and why these two goals (narrow main lobe, low sidelobes) are in tension for a fixed window length N . (Hint: think about the time-domain taper's smoothness versus its duration.)

(a) $W(e^{j\omega})$ acts like a smoothing kernel in the frequency domain. $W(e^{j\omega})$ with a narrow main lobe results in every point of $H_w(e^{j\omega})$ being a weighted average of $H_d(e^{j\omega})$ over a small neighborhood of frequencies. Thus, the cutoff at ω_c is only slightly blurred and the transition stays narrow.

A wide main lobe, conversely, averages over a larger frequency neighborhood, thus spreading the transition over a wider band.

(b) The sidelobes of W lead to leaking energy from the passband into the stopband (and vice versa) during the convolution with $H_d(e^{j\omega})$. Stopband frequencies pick up small contributions from the passband values of $H_d(e^{j\omega})$, weighted by the window's sidelobes.

Lower sidelobes mean less leakage.

(b) (contd.)

The tension arises because a window's time domain taper that goes smoothly to zero at its edges (and thus has lower sidelobes) is more more narrow in the time domain than a full-length rectangular window, leading to a wider main lobe in the frequency domain.

In general it is not possible to make the time domain taper very sudden at the time domain edges ($\Rightarrow$ narrow main lobe in frequency) and very smooth ($\Rightarrow$ low sidelobes in frequency) within the same finite window length.